Journal of Korean Institute of Intelligent Systems
Vol. 36, No. 5, October 2026, pp. 445-459
http://dx.doi.org/10.5391/JKIIS.2026.36.5.445

ISSN(Print) 1976-9172
ISSN(Online) 2288-2324

FinCat: 금융 이해력 교육을 위한 게이미피케이션 기반 멀티 에이전트 토론 시스템

FinCat: A Gamified Multi-Agent Debate System for Financial Literacy Education

정다윤†*, 이동현†*, 최윤선**, 강윤석***‡, 김보관****‡
Dayun Jeong†, Dong Hyeon Lee†, Yoonseon Choi, David Yoon Suk Kang‡, and Bogoan Kim‡

*충북대학교 정보통신공학부 학사과정, **한양대학교 연구원,
***충북대학교 소프트웨어학부 조교수, ****충북대학교 정보통신공학부 조교수
*B.S. Course, School of Information and Communication Engineering, Chungbuk National University
**Research Associate, Hanyang University
***Assistant Professor, School of Computer Science, Chungbuk National University
****Assistant Professor, School of Information and Communication Engineering, Chungbuk National University

요 약

청년층의 금융시장 참여는 늘지만 금융 이해력은 낮다. 금융 의사결정에는 정답이 없어, 교육은 정답을 주입하기보다 학습자가 여러 관점을 비교해 스스로 판단하도록 이끌어야 한다. 그러나 기존 게임형 학습은 흥미를 끌지만 개인화된 다관점 피드백이 부족하고, AI 자문 도구는 개인화돼도 하나의 권고로 수렴한다. 본 연구는 퀘스트 기반 시뮬레이션과 대규모 언어모델 기반 다중 에이전트 토론을 결합한 금융 교육 시스템 FinCat을 제안한다. 시뮬레이션은 검증된 금융 이해력 평가를 토대로 위험 없이 의사결정을 연습하게 한다. 토론에서는 투자 성향이 다른 세 에이전트와 심판이 경제 뉴스에 대한 학습자 판단에 반응한다. 성능 측정, 청년 20명 연구, 전문가 5명 인터뷰로 이를 평가하였다. 그 결과 시스템은 상호작용에 빠르게 반응하였고 사용성도 높았으며, 학습자와 전문가 모두 이 토론을 상반된 관점을 견주는 유익한 방식으로 평가하였다. 끝으로 FinCat 설계와 다중 에이전트 토론을 금융 교육 학습 활동으로 활용할 시사점을 제안한다.

키워드 : 금융 이해력 교육, 대규모 언어모델, 다중 에이전트 토론, 게이미피케이션

Abstract

Young adults increasingly participate in financial markets, yet often do so with limited financial literacy. Financial decisions are open-ended and rarely have a single correct answer, so financial education should help learners compare competing perspectives and form their own judgments rather than acquire prescribed ones. Existing tools address only part of this need. Gamified learning environments engage learners but seldom provide personalized, multi-perspective feedback, and AI-based advisory tools provide personalized feedback but are typically oriented toward a single recommendation. We present FinCat, a financial education system that pairs a quest-based simulation with a multi-agent debate module built on large language models. The simulation draws its content from a validated financial-literacy assessment and lets learners rehearse decisions without real risk. In the debate module, three agents with distinct investment personas and a Judge agent respond to the learner's own judgment about an economic news event. We evaluate FinCat by measuring its runtime performance, conducting a user study with 20 young adults, and interviewing five domain experts. The system responds quickly enough for interactive play, learners rate it as usable, and both learners and experts regard the multi-agent debate as a valuable way to reason about competing financial perspectives. This work contributes the design of FinCat and a set of implications for using multi-agent debate as a learning activity in financial education.

Key Words : Financial literacy education, large language models, multi-agent debate, Gamification

Received: Jul. 28, 2026
Revised : Aug. 21, 2026
Accepted: Aug. 24, 2026
†Equal contribution to this work
‡Corresponding author
(dyskang@cbnu.ac.kr, bogoan@cbnu.ac.kr)

본 논문은 정부(과학기술정보통신부)의 재원으로 한국연구재단의 지원을 받아 수행된 연구임(RS-2026-25490215, RS-2026-25601465, RS-2026-25494906).

This is an Open-Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/3.0) which permits unrestricted noncommercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

1. INTRODUCTION

Rising asset inequality, together with a perception that labor income alone cannot keep pace with asset appreciation, has led many young adults to participate in financial markets such as equities and cryptocurrencies. Such participation, however, is often motivated not by adequate financial knowledge but by psychosocial factors such as peer influence, information asymmetry, and fear of missing out[1-5]. Among young adults whose financial experience remains limited, this tendency can lead to irrational financial behaviors, including poor judgment and excessive leveraging[6,7].

A fundamental driver of such behavior is a lack of financial literacy[8], which encompasses the knowledge, attitudes, and behaviors necessary for sound financial decision-making[9]. According to the 2024 National Financial Literacy Survey conducted by the Bank of Korea and the Financial Supervisory Service, young adults showed relatively low levels of financial literacy compared with other age groups, excluding those in their 70s[10]. Young adulthood is a critical transition toward financial independence, during which lasting financial habits are formed. For this reason, the need for structured educational interventions is well recognized[11].

The challenge of financial education, however, does not lie in a lack of knowledge alone. Financial decision-making is inherently open-ended, as there is often no single correct answer, and the appropriate choice in a given situation depends on an individual's risk preference and context. The goal of financial education is therefore not to convey normative answers, but to help learners compare different perspectives and form their own judgments. Conventional financial education, nonetheless, tends to rely on instructor-led, unidirectional knowledge transfer, which offers learners few opportunities to practice decision-making and reflect on its consequences[12,13].

Two established technical approaches each address only part of this challenge. On one hand, gamified and simulation-based learning is known to promote active learning and engagement[14-17]. In practice, however, such environments often lack personalized feedback and rarely present multiple perspectives[18]. On the other hand, AI-based financial advisory tools provide personalized feedback, but their guidance is typically unidirectional and oriented toward a single recommendation[19]. This runs counter to a learning task whose aim is to convey that no single correct answer exists. AI-driven personalization has been shown to improve learning outcomes[20], yet environments that expose learners to diverse perspectives while allowing them to actively form and reconstruct their own financial judgments remain underdeveloped[21-23].

We frame this gap as a design problem: designing a learning environment that lets learners practice financial decision-making repeatedly in a low-stakes setting, while encouraging them to compare competing perspectives rather than adopt a single prescribed answer. Two design requirements follow. First, the environment needs a gamified simulation in which financial decisions can be rehearsed without real financial consequences. Second, it needs agents that can present different investment perspectives at once and respond to a learner's input in a context-sensitive manner. For the second requirement, conventional agent-based modeling approaches rely on predefined deterministic rules or fixed state machines, and thus cannot readily generate open-ended financial reasoning or respond to a learner's free-form opinions[24, 25]. Large language models (LLMs) can meet this requirement, as their natural-language reasoning and generation can be shaped by distinct personas. Assigning a different persona to each agent allows the system to provide multi-perspective feedback that extends beyond the reasoning of a single model[26].

In response to this design problem, we present *FinCat,* a financial education system that combines gamification with a multi-agent LLM architecture. *FinCat* couples a quest-based simulation with a debate module. The simulation is organized around the financial-behavior items on which young adults are most often found to be deficient. The debate module comprises three LLM agents of distinct investment personas-conservative, moderate, and aggressive-and a Judge agent that synthesizes their arguments. Rather than passively receiving unidirectional advice, the learner submits a personal judgment and receives targeted critique from agents that hold opposing perspectives. Financial matters are often regarded as socially sensitive and difficult to discuss with peers[27]. *FinCat* offers an alternative setting, in which learners examine their own financial judgments from multiple perspectives through interaction with the agents rather than with peers. We conducted a user study centered on four quests that cover the system's core learning activities, including the multi-agent debate. We examine the usability of

FinCat and its perceived educational value, and draw implications for the design of multi-perspective financial education systems.

This paper makes the following contributions:

- We articulate the design problem of supporting active, multi-perspective financial-judgment formation, and we present *FinCat*, a system that combines a quest-based financial simulation with an LLM-based multi-agent debate module in response.
- We propose a structured debate mechanism in which the learner participates actively rather than receiving advice passively. The learner submits an analysis, receives targeted persona-specific critique, and reconsiders their reasoning through a synthesized Judge report.
- We demonstrate, through a user study, that *FinCat* is usable and perceived as educationally valuable, and we discuss implications for designing multi-agent debate as a learning activity within an interactive game environment.

2. RELATED WORK

2.1 GAME-BASED AND GAMIFIED FINANCIAL EDUCATION

Prior work reports that game-based and gamified approaches can improve engagement and learning outcomes across educational domains[28], including financial education[18,29]. Moonshot[30] uses a board-game format to raise students' interest in finance, and Oma Onni[18] provides a gamified online environment with points and badges, though neither incorporates AI-based feedback or personalization. FinCraft[21] adds archetype-based personalization through a serious game, but its feedback is not driven by interactive AI agents. Mabu[22] combines gamification with goal-directed design and offers goal-based personalization, yet presents a single perspective rather than several. John and the Rainbow Coins[23] integrates multiple AI agents to provide context-aware financial advice within a game. However, it primarily delivers a single synthesized recommendation, limiting exposure to competing perspectives and leaving little room for learners to present their own judgments. *FinCat* addresses this limitation by assigning conflicting personas to its agents, requiring learners to actively evaluate opposing viewpoints and form independent judgments. Table 1 summarizes how these tools compare with *FinCat* across five characteristics.

Beyond game-based tools, AI-based financial advisory systems have been studied as a way to deliver personalized guidance. Recent work, however, finds that such systems tend to produce recommendations for the user rather than prompt the user to form and examine their own judgment[19,31]. Across both lines of work, learners remain largely recipients. Few tools combine the engagement of gamified practice with feedback that is personalized, presents more than one perspective, and invites learners to reason for themselves. *FinCat* is designed to fill this gap by pairing a gamified financial simulation with a multi-agent debate in which learners actively present and reconsider their own judgments.

2.2 LLM-BASED MULTI-AGENT SYSTEMS

Recent work has explored multi-agent systems built on LLMs, in which several agents take on

표 1. *FinCat*과 기존 금융교육 도구 비교 (JRC는 John and the Rainbow Coins을 의미)
Table 1. Comparison of *FinCat* with existing financial education tools (JRC refers to John and the Rainbow Coins)

	Moonshot [30]	Oma Onni [18]	FinCraft [21]	Mabu [22]	JRC [23]	*FinCat*
Gamification	Board game	Gamified online	Points, badges	Badges, tracker	Exploration & reward	Quest & reward
AI feedback	None	None	None	None	Quality-diversity -based multi-agent LLM	Multi-agent LLM
Personalization	None	None	Archetype-based	Goal-based	Behavior-based	Tendency-based
Multiple perspectives	None	None	None	None	None	Conservative, moderate, aggressive
Platform	Offline	Web	Web	Mobile	Web	Desktop

different roles and interact to address tasks that exceed the reach of a single model[25,32,33]. Assigning a distinct persona to each agent broadens the system's reasoning and surfaces a wider range of considerations than a single prompt[26,34]. Such architectures have supported analytical tasks across domains, from design ideation[35] to statistical reasoning [36], and have been applied to financial settings such as portfolio construction, where several agents jointly analyze market conditions[37]. LLM-based systems have also been studied as interactive partners in educational settings[38,39]. A recurring element in these systems is an adjudicating agent that consolidates the others' outputs; adding such a step has been reported to improve the consistency of the final result and to reduce unreliable content in LLM-generated text[32].

These systems, however, are designed largely so that the agents reason among themselves or on the user's behalf, leaving the human outside the deliberation. Conventional agent-based modeling frameworks, in turn, rely on predefined deterministic rules or fixed state machines[24,25], and cannot readily produce the open-ended reasoning or context-sensitive responses that an interactive learning setting calls for. Placing a human inside such a deliberation, so that they actively contribute and revise their own reasoning, has been explored in other educational contexts. For example, recent work[38] has investigated how users interact with LLM-based conversational AI not merely as a static information retrieval tool, but as a collaborative partner to iteratively reflect on and refine their decisions. However, while the educational value of such human-in-the-loop deliberation is increasingly recognized, its application remains largely under-explored in financial education. Specifically, there remains a lack of systems designed to support learners in presenting their own opinions first, exchanging multi-faceted feedback with agents holding opposing perspectives, and independently reflecting on their financial decision-making process.

3. FINCAT

FinCat is a financial education system in which learners practice financial decision-making through quests and examine economic news through a debate with multiple LLM agents. We describe the reasoning behind its design, the architecture that supports it, and the two components that learners engage with: a quest-based simulation and a multi-agent debate.

3.1 DESIGN RATIONALE

The design of *FinCat* reflects how we answered three questions raised by the gaps discussed in Section 2: what learners should practice, how they should practice it, and how they should receive feedback.

For what learners should practice, we grounded the content in a validated assessment rather than in topics chosen ad hoc. We took the financial-behavior items of the 2024 National Financial Literacy Survey conducted by the Bank of Korea and the Financial Supervisory Service as the learning targets, since young adults score low on financial literacy relative to most other age groups[10], consistent with prior findings that young adults' financial literacy remains persistently low[40]. From these items, we selected four behavior areas that represent core financial behaviors covered by the survey: assessing one's own financial situation, managing credit, selecting a financial product on an informed basis, and analyzing economic information before making a decision. We translated each area into a quest (Section 3.3), so that the learning content stays tied to established dimensions of financial literacy rather than to arbitrary scenarios, while keeping the set of quests to a manageable length.

For how they should practice, we situated the content in a gamified simulation rather than in a sequence of expository lessons. Because habits form through repeated decisions and self-regulation over time[41], we let learners make choices that accumulate across turns and observe their outcomes within the game, so that practice carries no real financial risk.

For how they should receive feedback, we organized it as a debate among agents rather than a single recommendation. Because a financial decision rarely has one correct answer, a single piece of advice would misrepresent the nature of the task. We therefore expose learners to agents of differing investment orientations and ask them to weigh competing positions while forming their own judgment.

FinCat draws on serious-game research but is best characterized as a gamified system, since it applies game design elements—such as quests and collectible cards—within a non-game context to sustain learner engagement and stimulate intrinsic motivation[17]. Specifically, *FinCat* targets three core motivational drivers: autonomy (through open-ended investment choices and debate participation), competence (via immediate feedback and rewards

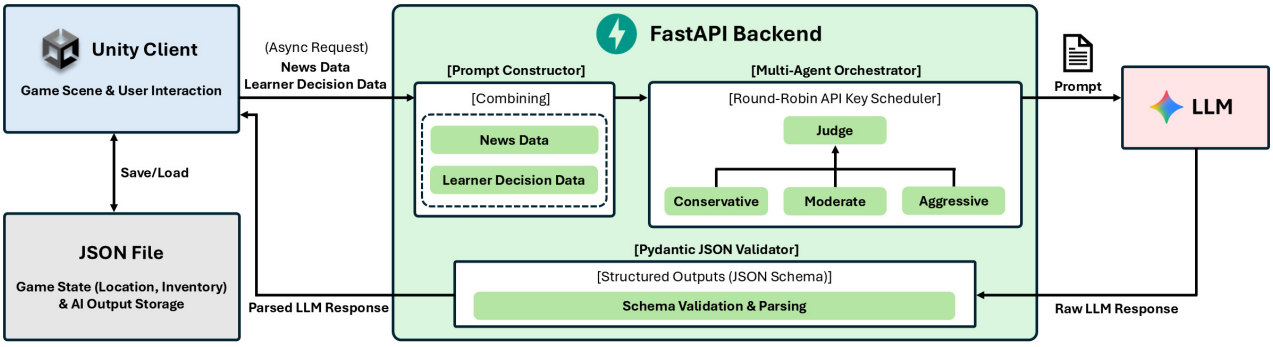


그림 1. *FinCat*의 3계층 아키텍처 및 데이터 흐름
Fig. 1. Three-tier architecture and data flow of *FinCat*

from quests and card collection), and relatedness (by interacting with distinct persona agents). Thus, these gamification mechanisms are intended not merely for entertainment, but as active drivers of educational motivation.

3.2 SYSTEM ARCHITECTURE

FinCat is built on a three-tier architecture comprising a Unity game client, a FastAPI backend server, and the Google Gemini 2.5 Flash Lite API. The overall orchestration and data flow are shown in Fig. 1. When a user action occurs, the Unity client sends a REST API request in JSON format to the backend, which constructs a prompt from the request and forwards it to the Gemini API. The response is parsed against a predefined JSON Schema and rendered on the client in real time. Game state, including the player's location, inventory, and stored agent outputs, is persisted on the client as a JSON file. All simulation-related computations, including stock price updates, bank interest accrual, and the calculation of interest income tax, transaction tax, and capital gains tax, are performed on the Unity client. In contrast, the backend handles AI-related operations, including prompt constructions, multi-agent orchestration, and agent response generation. The system is structured in this way to separate game simulation logic from AI-related processing, allowing the two components to be developed, modified, and maintained independently. This separation also enables multi-agent functionality to be integrated and extended without modifying the core game mechanics.

We introduced three techniques to keep this pipeline stable in an interactive game setting. First, structured outputs are enforced with Pydantic so that every LLM response conforms to a strict JSON Schema, which removes parsing errors. Second, a round-robin scheduler distributes requests across multiple API keys, which mitigates the rate limits that arise when several agents are invoked at once. Third, asynchronous communication is implemented with Python asyncio on the backend and the UniTask library on the Unity client, so that neither the game loop nor the server thread is blocked during inference. For concurrency, we follow the engineering rationale of NL4Py[42] and adapt it to the latency of LLM inference.

3.3 QUEST-BASED SIMULATION

In *FinCat*, the learner plays as a character whose goal is to accumulate 100 million KRW to purchase a target item. The game world is a single village of six functional zones(Fig. 2), summarized in Table 2. The player advances the game in discrete turns, and each turn updates the player's assets, so that the consequences of earlier decisions accumulate and become visible over time.



그림 2. *FinCat* 게임 월드의 6개 기능 구역
Fig. 2. Six functional zones of the *FinCat* game world

Before entering the game world, the learner completes a brief assessment to identify their personal investment tendency (e.g., risk-averse or risk-seeking). Rather than mechanically restricting in-game content based on this profile, the system utilizes it as a cognitive scaffold, encouraging learners to reflect on their

tendency when evaluating the agents' advice later in the game.

The game includes four quests, each corresponding to one of the four behavior areas identified in Section 3.1 and set in the zone where that behavior naturally occurs (Table 3). Each quest opens with a short explanation of the relevant concept, so that the learner acquires the necessary background before making a decision. Completing a quest awards a collectible card, and collecting all four cards completes the quest sequence. The first three quests use structured formats such as selection, matching, and a quiz, while the fourth, set in the Debate Hall, has the learner analyze an economic news event and form a judgment through discussion with the agents.

표 2. *FinCat* 게임 월드의 구역별 기능
Table 2. Functions of each zone in the *FinCat* game world

Zone	Function
Home	Advance turns by interacting with the bed object
Bank	Manage assets through savings, time deposits, and checking accounts
Brokerage	Trade investment products such as stocks and bonds
School	Review financial concepts through question-and-answer sessions with the agents
Debate Hall	Analyze economic news and debate with the agents
Shop	Purchase the target item required to complete the game

표 3. 4개의 퀘스트 구성
Table 3. Structure of the four quests

Order	Zone	Behavior Area and Activity
1	School	Assessing one's financial situation; select the option closest to one's own circumstances
2	Bank	Managing credit; term-description matching and a true/false quiz
3	Brokerage	Informed product selection; choose the most appropriate product for a scenario
4	Debate Hall	Economic analysis; review agent opinions, submit an opinion, and receive feedback

The player advances a turn by interacting with the bed object in the Home zone, which triggers, in order, the accrual of bank interest, the payment of salary, and the update of market prices; all changes are reflected in the asset module in real time. Stock prices update each turn according to (1), in which a news-dependent impact term drives the price and a small random fluctuation is added:

$$P_{t+1} = round(P_t(1+\alpha+\epsilon)), \epsilon \sim U(-0.01, 0.01) \quad (1)$$

where P_t is the price at turn t, α is a news-dependent impact coefficient defined for each stock, and ϵ is uniform random noise that introduces a small fluctuation to represent short-term market variability not directly determined by the selected news event. The round($\cdot$) operator rounds the calculated stock price to the nearest integer. The news-dependent impact coefficient α was determined based on observed stock-price responses to relevant news events. Stocks in the game were categorized by industry, and a representative company with the highest market capitalization within each industry was selected as a reference, yielding five stocks in total. Historical stock-price data for these representative companies were reviewed to identify specific cases in which a clear relationship between a news event and a subsequent stock-price movement was observed. The direction and magnitude of the observed price movement were then used as a basis for assigning the corresponding α value to the relevant stock and news event. The values of α were set within the range of –0.29 to 0.29, allowing the uniform random noise ϵ, which ranges from –0.01 to 0.01, to be incorporated while keeping the resulting stock-price change within –30% to 30%. This ±30% limit was defined with reference to the ±30% daily price limit applied to Korean-listed stocks.

Bonds are modeled as a more stable asset class: their price converges gradually toward the face value as the remaining maturity decreases, with small fluctuations along the way. At maturity, after-tax interest is settled by deducting an interest income tax of 15.4%, and when a stock is sold, a transaction tax of 0.2% and a capital gains tax of 22% are applied to determine the net proceeds. These tax rates are simplified assumptions adopted to support the financial education objectives of the simulation and do not fully reflect the actual taxation rules for Korea-listed stocks.

The banking system supports three account types. A checking account allows immediate deposits and withdrawals but yields no interest. A savings account accrues simple interest each turn at a fixed annual

rate divided by the number of turns per year. A time deposit offers a higher rate but imposes a penalty for early withdrawal, and credits compound interest to the checking account at maturity. A fixed salary is paid to the checking account each turn, giving the learner a steady income to manage across turns.

3.4 MULTI-AGENT DEBATE

The fourth quest takes place in the Debate Hall, where the learner analyzes an economic news event together with a set of LLM agents. Unlike conventional agent-based modeling frameworks such as Mesa [24] and Evoplex [25], whose agents follow predefined rules or fixed state machines, *FinCat* uses LLM-based agents whose responses are shaped by persona-specific prompts.

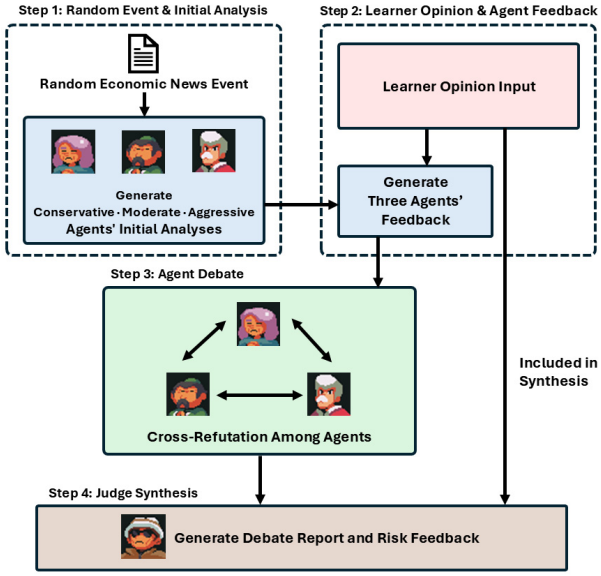


그림 3. 다중 에이전트 토론의 운영 파이프라인
Fig. 3. Operational pipeline of the multi-agent debate

In *FinCat*, each agent is assigned a distinct investment persona, so that the debate surfaces conservative, moderate, and aggressive readings of the same news event rather than a single analysis.

The module comprises four agents. The conservative agent emphasizes risk factors, forecasts prices cautiously, and favors deposit-oriented strategies. The moderate agent weighs macroeconomic indicators and market equilibrium to produce balanced forecasts. The aggressive agent emphasizes growth opportunities, forecasts optimistically, and favors active equity purchases. The Judge agent then synthesizes the three agents' forecasts and arguments together with the learner's own opinion into a single report. We introduced the Judge as a dedicated adjudicating step because consolidating multiple agent outputs has been reported to improve the consistency of the final result and to reduce unreliable content in LLM-generated text[32].

The debate proceeds through the stages shown in Fig. 3. When the learner enters the Debate Hall and approaches the Judge, an introductory screen outlines the workflow(Fig. 4(a)). The server then draws one of five economic news events, and the three persona agents each generate an independent analysis and forecast, which are presented to the learner(Fig. 4(b)). After reviewing these analyses, the learner enters a personal investment opinion. Each agent reviews this opinion, identifies gaps in the learner's reasoning, and returns tailored feedback(Fig. 4(c)). When the learner confirms the result, the agents critique one another's arguments in an internal cross-refutation step. The Judge agent then summarizes the discussion, assesses the weaknesses in the learner's reasoning, and outputs a structured report that is rendered on the client in real time(Fig. 4(d)).

To keep the agents' personas consistent and their output readable within the game interface, we applied three prompt-engineering techniques. First, persona-based chain-of-thought prompting assigns each agent a fixed investment persona and has it reason step by step through four indicators-interest rate, liquidity, risk, and fees-before producing a forecast, so that each output reflects the assigned perspective in a traceable manner. Second, constraint engineering enforces output rules suited to the Unity speech-bubble interface: responses are limited to two lines(about 120 characters), and agents may not repeat their role names or use conversational prefixes, which prevents interface overflow. Third, few-shot prompting supplies example input-output pairs of structured news data and the expected persona-specific analysis, which anchors the output format and reduces deviation from the schema. Because each persona is defined in a separate prompt file and every agent shares the same structured input-output interface, a new persona can be added without modifying the debate pipeline.

The final report is produced as a Pydantic schema-based structured output, with a rule appended to the prompt that requires pure JSON without Markdown. The output object conforms to a validation schema covering the persona-specific analyses and the debate fields, so that the cross-refutation summary and the assessment of the learner's reasoning are

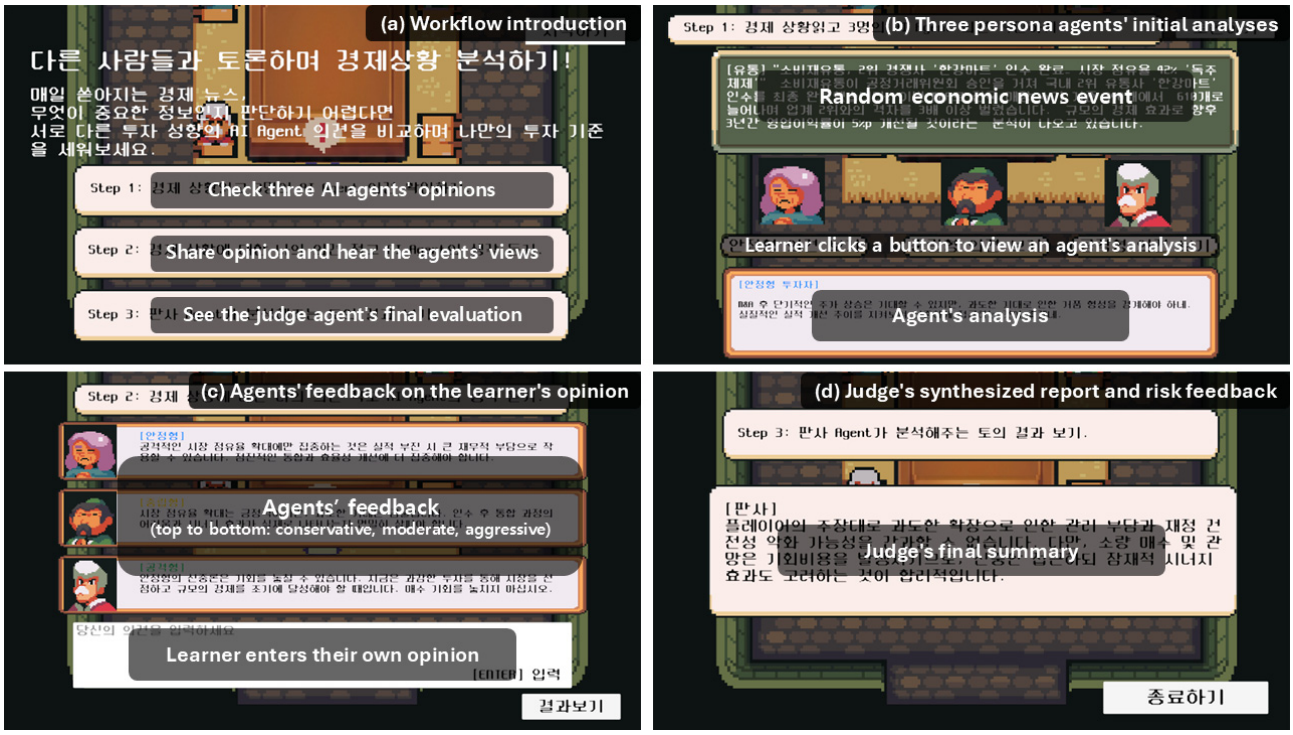


그림 4. *FinCat*의 다중 에이전트 토론 인터페이스: (a) 토론 절차 안내; (b) 세 페르소나 에이전트의 초기 분석; (c) 학습자 의견을 제출; (d) Judge 에이전트의 최종 보고서
Fig. 4. Multi-agent debate interface in *FinCat*: (a) the workflow introduction; (b) the three persona agents' initial analyses; (c) the learner submitting an opinion; and (d) the Judge's final report

transmitted to the interface without parsing errors.

Because *FinCat* is intended for learners who are still developing their financial literacy, we treat the reliability of its output as a design concern rather than an afterthought. The system presents a persistent notice clarifying that all agent output is generated for educational purposes and does not constitute investment advice, and the debate is framed throughout as a means of comparing perspectives rather than as a source of actionable recommendations.

4. EVALUATION

We evaluated *FinCat* along three dimensions of its feasibility as an educational system: whether the multi-agent pipeline runs fast enough to support real-time interaction, whether learners find the system usable and educationally valuable, and whether domain experts judge its design and output to be appropriate. We report these in turn as system performance, a user study, and an expert evaluation.

4.1 SYSTEM PERFORMANCE

All measurements were taken on a machine with an AMD Ryzen 7 4700U CPU, 16 GB of RAM, and integrated AMD Radeon graphics, running Windows 11 (64-bit). The backend ran on Python 3.10 with FastAPI, and the Gemini 2.5 Flash Lite API served as the inference engine. We repeated each stage of the debate pipeline ten times and recorded the response latency; the results are summarized in Table 4.

A complete debate session took on average 5.779 s. At this speed, the debate does not noticeably interrupt gameplay, because the asynchronous design keeps the game loop responsive during inference. The Dialogue stage (corresponding to Steps 1 and 2 in Fig. 3) took an average of 2.767 s. In this stage, to

표 4. *FinCat* 토론 파이프라인의 지연시간 (10회 반복 실험의 평균)
Table 4. Latency of the *FinCat* debate pipeline (averaged over 10 trials)

Stage	Agents	Latency (s)
Dialogue 1 (Step 1-2)	Conservative, moderate, aggressive	2.767
Refute & Judge (Step 3-4)	Judge final report	3.012
Total		5.779

optimize latency, the system utilizes a single API call where the LLM is prompted to generate logically independent chain-of-thought reasoning for all three persona agents simultaneously, outputting a consolidated and validated JSON. During this stage, the average prompt length was about 207 input tokens, and the model generated about 104 output tokens per agent.

The subsequent Refute & Judge stage (corresponding to Steps 3 and 4 in Fig. 3) accounted for a larger portion of the total latency, requiring 3.012 s. In this stage, the accumulated dialogue increased the input context significantly to about 1,453 tokens. To optimize system responsiveness, the agents' refutations on the learner's opinion and the Judge's final report are not processed sequentially. Instead, they are generated simultaneously through a single consolidated API call. This architectural choice explains why Table 4 reports a single combined latency for these two logical steps. Although the latency was higher than in the Dialogue stage, the increase was kept minimal thanks to this consolidated prompt design. The standard deviation of the total latency across the ten trials was 0.85 s, and no JSON parsing failures occurred, consistent with the schema-enforced structured output.

4.2 USER STUDY

4.2.1 Participants

We recruited 20 young adults in their twenties(15 male and 5 female) through an online messaging platform. Eligible participants were adults aged 19 or older who understood the study and voluntarily consented to take part, and who had no difficulty completing the questionnaires or communicating during the session. This study was approved by the Institutional Review Board (IRB) of Chungbuk National University (Approval No. CBNU-2026-A-0150). All participants provided informed consent before the session.

4.2.2 Procedure

The study was conducted in person in a laboratory at the authors' institution, and each session took about 40 minutes. After the researcher explained the purpose, procedure, data collected, and privacy and anonymization policy, the participant gave informed consent. The participant then played *FinCat* on a laboratory PC for about 10 to 20 minutes, proceeding through the introductory story, an interaction tutorial, and the four quests in sequence, including the multi-agent debate in the Debate Hall. We recorded the screen to capture decision choices, play time, and progression patterns. After the session, the participant completed a set of post-session questionnaires and answered open-ended interview questions, which took about 20 minutes. Beyond the in-game instructions, we provided no separate training, so the session reflected first-time use of the system.

4.2.3 Measures

We assessed *FinCat* with three sets of post-session measures. First, we measured usability with the System Usability Scale(SUS) [43] and perceived workload with an adapted version of the NASA Task Load Index (NASA-TLX) [44], in which some items were phrased positively (e.g., perceived performance and the adequacy of the available time) so that, for these items, higher scores indicate a more favorable experience. Second, to capture perceived educational value, we conducted a post-session interview asking participants how helpful they found the system as a whole and each of its main features (the tutorial quests, the financial simulation, and the multi-agent debate), along with their self-assessed learning, decision-making confidence, and satisfaction. Third, we administered items adapted from the 2024 National Financial Literacy Survey conducted by the Bank of Korea and the Financial Supervisory Service questionnaire [10], covering the knowledge, attitude, and behavior dimensions of financial literacy; because these items were collected only once, after the session, we report them to describe participants' self-reported standing across the three dimensions rather than as a measure of learning gains.

4.2.4 Results

All 20 participants completed the four quests,

표 5. 조정된 NASA-TLX 하위 척도 점수(N = 20, Likert 7점 척도) 요구도 관련 하위 척도는 낮을수록(↓), 긍정 문항으로 구성된 하위 척도는 높을수록(↑) 바람직
Table 5. Adapted NASA-TLX subscale scores(N = 20, 7-Point Likert scale) Lower is more favorable for demand-related subscales (↓), Higher is more favorable for positively phrased subscales (↑)

Subscale	Mean	SD
Mental demand(↓)	4.05	1.43
Physical demand(↓)	2.50	1.85
Effort(↓)	2.80	1.40
Frustration(↓)	2.80	1.82
Performance(↑)	5.47	0.87
Temporal adequacy(↑)	5.60	1.27

including the multi-agent debate, within the session. Participants rated the usability of *FinCat* positively, with a mean SUS score of 73.75, which falls in the "Good" range[45]. On the adapted NASA-TLX, the demand-related subscales (mental demand, physical demand, effort, and frustration) received low scores, while the positively phrased subscales (performance and the adequacy of the available time) received high scores; the subscale scores are reported in Table 5. Together, the two measures indicate that participants found *FinCat* usable and completed the tasks without excessive perceived workload.

Turning to perceived educational value, participants reported that *FinCat* was helpful for engaging with financial concepts, and identified the multi-agent debate as the most helpful feature. One participant noted, "*Seeing the conservative and aggressive agents disagree made me realize there wasn't one right answer, so I started thinking about which view fit my own situation*" (P3). Another remarked that the Judge's summary "*helped me see what I had missed in my own reasoning*" (P17). These responses suggest that the debate module, the central design element of *FinCat*, was perceived as supporting learners' engagement with multiple perspectives.

Finally, on the items adapted from the 2024 National Financial Literacy Survey administered after the session, participants reported generally positive levels across the three dimensions (knowledge: 3.83 out of 5; attitude: 3.51 out of 5; behavior: 0.88 out of 1). We report these values only to characterize the participants, and do not interpret them as evidence of learning, since the study included neither a pre-session baseline nor a comparison condition.

4.3 EXPERT EVALUATION

4.3.1 Participants

To examine the validity of the design and the appropriateness of the agent-generated output, we conducted semi-structured interviews with five domain experts: a financial-literacy researcher (E1), a private banker at a securities firm (E2), a certified financial planner (E3), and two university professors in finance (E4 and E5). The financial planner had more than five years of professional experience, and the others each had more than ten years. Instead of interacting with the system directly, the experts evaluated *FinCat* by observing recorded video sessions of the gameplay.

4.3.2 Procedure and Analysis

Prior to each interview, we provided the experts with recordings of three participants' sessions from the user study. Although all participants went through the same system, we selected three whose investment orientations differed, so that the experts could observe the range of interactions the system produces across conservative, moderate, and aggressive profiles.

Each interview lasted on average 47 minutes and addressed four topics: the validity of the quest design, the educational value of the multi-agent debate, the appropriateness and reliability of the agent-generated output, and the system's applicability and limitations, as summarized in Table 6. All interviews were audio-recorded and transcribed.

표 6. 전문가 평가를 위한 인터뷰 주제 및 예시 질문
Table 6. Interview topics and sample questions for the expert evaluation

Topic	Sample Question
Quest design validity	Do the tutorial quests appropriately reflect the financial behaviors that young adults need to develop?
Value of the debate	How well does the multi-agent debate, rather than a single recommendation, suit the way financial decisions are made?
Output appropriateness and safety	Are the agent-generated analyses sound, and are the educational framing and disclaimer adequate for young learners?
Applicability and limitations	In which educational settings could the system be used, and what are its main risks or weaknesses?

Two of the authors independently open-coded the full set of transcripts, and a third, more senior author reviewed the coding and adjudicated disagreements. Agreement between the two coders was assessed with Cohen's κ, computed separately for each of the eight codes; agreement averaged 0.83(range 0.77–0.91), indicating almost perfect agreement [46].

4.3.3 Findings

Quest design validity. The experts regarded the grounding of the tutorial quests in the financial behavior areas of the 2024 National Financial Literacy Survey as appropriate for young learners. The financial-literacy researcher considered the choice of behaviors well justified: "*Targeting these four behaviors is very reasonable. They are essential considerations*

in financial-literacy education, and they are exactly where young adults struggle" (E1).

Value of the multi-agent debate. The experts viewed the debate structure, in which learners weigh competing perspectives rather than receive a single recommendation, as well matched to the open-ended nature of financial decision-making. One professor observed, "*Presenting conflicting views rather than a single answer is closer to how financial decisions are actually made, and it lets learners think for themselves instead of just accepting a conclusion*" (E5).

Appropriateness and reliability of agent output. The experts from professional practice, the private banker and the certified financial planner, judged the agent-generated analyses to be sound overall. The private banker noted that the aggressive agent occasionally phrased its position more assertively than is suitable for an educational setting (E2). The certified financial planner assessed the output positively but suggested that the system could also convey the downside risks of investing, not only its opportunities: "*The analyses were reasonable, but investing has a risky side as well, and it would be good to convey that. For young users especially, the system should keep making clear that this is not investment advice*" (E3).

Applicability and limitations. Both professors saw *FinCat* as applicable to classroom use and self-directed learning alike, and pointed to the limited amount of content as its main constraint. One professor noted, "*This would fit well as a hands-on component in an introductory finance course, though the content is still limited, for example the number of news events, so learners would need additional material to go deeper*" (E4).

5. DISCUSSION

5.1 DESIGNING MULTI-AGENT DEBATE AS A LEARNING ACTIVITY

Across the user study and the expert evaluation, the multi-agent debate, the central design element of *FinCat,* was consistently identified as the most valuable part of the system (Sections 4.2 and 4.3). Both learners and experts valued it for the same reason: it presents competing positions rather than a single answer, which learners connected to the realization that no one answer is correct and which experts saw as matching the open-ended nature of financial decisions. This convergence supports the design decision behind *FinCat*, that feedback in financial education can be organized as a debate among differing positions rather than as a single recommendation. Taken together with the system's runtime performance and its usability (Sections 4.1 and 4.2), it indicates that the design is both practical to build and well received. Our evaluation concerns the feasibility and reception of this design; it does not establish whether the system improves financial literacy.

This reading also informs how we view the underlying simulation. As described in Section 3.3, the market and banking models in *FinCat* are deliberately simplified, and are meant to give learners a legible setting in which the consequences of decisions accumulate and become visible, rather than to reproduce the dynamics of real markets. We regard this simplification as appropriate for a tool whose purpose is reasoning practice rather than market fidelity.

5.2 EXTENSIBILITY AS A RESEARCH AFFORDANCE

A recurring theme in the expert interviews was the wish for a broader range of scenarios and more advanced content for repeated use (Section 4.3). *FinCat* is well positioned to accommodate such extension, because it separates content from logic: quests are defined as JSON files, agent personas as prompt files, and the language model is accessed through a replaceable endpoint, so the system can be reconfigured without changing its core code. This modularity also makes *FinCat* a useful instrument for future research. Researchers can edit quest and persona files to construct controlled comparisons, for example contrasting single-perspective and multi-perspective feedback, or substitute alternative language models to study how model choice affects the quality of the generated feedback. *FinCat* can thus serve not only as an educational tool but also as a platform for studying the questions its own design raises.

5.3 LIMITATIONS AND FUTURE WORK

Our evaluation was designed to assess the feasibility of FinCat, in terms of its usability and perceived value, rather than its effect on learning. Additionally, because some NASA-TLX items were rephrased with positive polarity (Section 4.2.3), the resulting workload scores are not directly comparable to published norms or to studies using the standard NASA-TLX.

They should therefore be interpreted as an index of participants' perceived workload within this study. As noted in Section 4.2, the study did not include a pre-session baseline or a comparison condition, and the financial-literacy items describe participants' self-reported standing rather than measured gains. The sample was also recruited via convenience sampling through an online platform and was skewed toward male participants(15 of 20), and all sessions were conducted in a laboratory at the authors' institution, which may have introduced response bias. Establishing whether *FinCat* improves financial literacy, and whether the multi-agent debate does so more effectively than single-source feedback, will require controlled and longitudinal studies, which the extensibility of the system is intended to support.

We also did not systematically assess the factual accuracy of the agent-generated analyses. We addressed output reliability in part through the expert review reported in Section 4.3 and through a persistent disclaimer that frames the output as educational rather than advisory, but a larger-scale evaluation of accuracy remains future work. Relatedly, *FinCat* currently relies on a single commercial language model(Gemini 2.5 Flash Lite). While we used this capable model to establish initial feasibility, the reasoning quality, persona adherence, and persuasive depth of the agents may vary depending on the underlying model. Therefore, a systematic comparison across different models to evaluate their impact on the debate quality and learner experience is left for future work, which the system's replaceable model interface is designed to facilitate. Finally, *FinCat* was developed and evaluated in a Korean-language context, and examining how it transfers to other languages and educational settings is a further direction for future work.

6. CONCLUSION

This paper presented *FinCat*, a financial education system that combines a quest-based simulation with a multi-agent debate in which learners form and revise their own financial judgments. Because financial decisions rarely have a single correct answer, we designed the system so that learners practice decisions in a low-stakes setting and receive feedback as a debate among agents of differing investment orientations rather than as a single recommendation. We evaluated *FinCat* by measuring its runtime performance, conducting a user study, and interviewing five domain experts. The system responds quickly enough for interactive play and was rated as usable, and both learners and experts saw the multi-agent debate as its most valuable element for reasoning about competing financial views. These results establish the feasibility of the design. Together with the system's modular structure, they provide a foundation for future studies of how multi-perspective feedback shapes financial learning.

Conflict of Interest

저자는 본 논문에 관련된 어떠한 잠재적인 이해상충도 없음을 선언한다.

References

[1] K. T. C. Nguyen, et al., "Fear of missing out and Generation Z's investment behavior in Vietnam's stock market: The moderating role of financial literacy," *Edelweiss Applied Science and Technology*, vol. 10, no. 1, pp. 500-510, https://doi.org/10.55214/2576-8484.v10i1.11624, 2026.

[2] M. Argan, V. Altundal, and M. Tokay Argan, "What Is the Role of FoMO in Individual Investment Behavior? The Relationship among FoMO, Involvement, Engagement, and Satisfaction," *Journal of East-West Business*, vol. 29, no. 1, pp. 69-96, https://doi.org/10.1080/10669868. 2022.2141941, 2023.

[3] N. W. S. Martaningrat and Y. Kurniawan, "The Impact of Financial Influencers, Social Influencers, and FOMO Economy on the Decision-Making of Investment on Millennial Generation and Gen Z of Indonesia," *Journal of Ecohumanism*, vol. 3, no. 3, pp. 1319-1335, https://doi.org/10.62754/joe.v3i3.3604, 2024.

[4] A. K. Przybylski, K. Murayama, C. R. DeHaan, and V. Gladwell, "Motivational, emotional, and behavioral correlates of fear of missing out," *Computers in Human Behavior*, vol. 29, no. 4, pp. 1841-1848, https://doi.org/10.1016/j.chb.2013.02.014, 2013.

[5] M. Nourallah, P. hman, and M. Amin, "No trust, no use: how young retail investors build initial trust in financial robo-advisors," *Journal of Financial Reporting and Accounting*, vol. 21, no. 1, pp. 60-82, https://doi.org/10.1108/jfra-12-2021-0451, 2023.

[6] E. Sinnewe and G. Nicholson, "Healthy financial habits in young adults: An exploratory study of the relationship between subjective financial literacy, engagement with finances, and financial decision-making," *Journal of Consumer Affairs*, vol. 57, no. 1, pp. 564-592, https://doi.org/10.1111/joca.12512, 2023.

[7] M. Van Rooij, A. Lusardi, and R. Alessie, "Financial literacy and stock market participation," *Journal of*

Financial economics, vol. 101, no. 2, pp. 449-472, https://doi.org/10.1016/j.jfineco.2011.03.006, 2011.

[8] A. Lusardi, "Financial literacy and the need for financial education: evidence and implications," *Swiss journal of economics and statistics*, vol. 155, no. 1, art no. 1, https://doi.org/10.1186/s41937-019-0027-5, 2019.

[9] A. Lusardi and O. S. Mitchell, "The Economic Importance of Financial Literacy: Theory and Evidence," *Journal of economic literature*, vol. 52, no. 1, pp. 5-44, https://doi.org/10.1257/jel.52.1.5, 2014.

[10] Bank of Korea and Financial Supervisory Service, "Results of the 2024 National Financial Literacy Survey," Available: https://www.bok.or.kr/portal/bbs/B0000502/view.do?menuNo=201265&nttId=10091152, 2025, [Accessed: Aug 14, 2026].

[11] C. Urban, M. Schmeiser, J. M. Collins, and A. Brown, "The effects of high school personal financial education policies on financial behavior," *Economics of education review*, vol. 78, art no. 101786, https://doi.org/10.1016/j.econedurev.2018.03.006, 2020.

[12] D. Fernandes, J. G. Lynch Jr, and R. G. Netemeyer, "Financial Literacy, Financial Education, and Downstream Financial Behaviors," *Management science*, vol. 60, no. 8, pp. 1861-1883, https://doi.org/10.1287/mnsc.2013.1849, 2014.

[13] A. Amagir, W. Groot, H. Maassen van den Brink, and A. Wilschut, "A review of financial-literacy education programs for children and adolescents," *Citizenship, Social and Economics Education*, vol. 17, no. 1, pp. 56-80, https://doi.org/10.1177/2047173417719555, 2018.

[14] T. Kaiser and L. Menkhoff, "Active learning improves financial education: Experimental evidence from Uganda," *Journal of Development Economics*, vol. 157, art no. 102870, https://doi.org/10.1016/j.jdeveco.2022.102870, 2022.

[15] J. Hamari, "Do badges increase user activity? A field experiment on the effects of gamification," *Computers in human behavior*, vol. 71, pp. 469-478, https://doi.org/10.1016/j.chb.2015.03.036, 2017.

[16] B. Kim, D. Jeong, H. Hong, and K. Han, "Narrating Routines through Game Dynamics: Impact of a Gamified Routine Management App for Autistic Individuals," *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, Honolulu, USA, pp. 1-15, https://doi.org/10.1145/3613904.3642357, May, 2024.

[17] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: defining "gamification"," *Proceedings of the 15th international academic MindTrek conference: Envisioning future media environments*, Tampere, Finland, pp. 9-15, https://doi.org/10.1145/2181037.2181040, Sept., 2011.

[18] P. Kalmi and J. Rahko, "The effects of game-based financial education: New survey evidence from lower-secondary school students in Finland," *The Journal of Economic Education*, vol. 53, no. 2, pp. 109-125, https://doi.org/10.1080/00220485.2022.2038320, 2022.

[19] T. Takayanagi, K. Izumi, J. Sanz-Cruzado, R. McCreadie, and I. Ounis, "Are Generative AI Agents Effective Personalized Financial Advisors?" *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, Padua, Italy, pp. 286-295, https://doi.org/10.1145/3726302.3729897, July, 2025.

[20] J. A. Kulik and J. D. Fletcher, "Effectiveness of Intelligent Tutoring Systems: A Meta-Analytic Review," *Review of educational research*, vol. 86, no. 1, pp. 42-78, https://doi.org/10.3102/0034654315581420, 2016.

[21] A. Rasco, J. Chan, G. Peko, and D. Sundaram, "Fincraft: Immersive Personalised Persuasive Serious Games for Financial Literacy Among Young Decision-Makers," *Proceedings of the 53rd Hawaii International Conference on System Sciences*, Maui, Hawaii, USA, pp. 32-41, Jan., 2020.

[22] P. L. To, "Enhancing Financial Literacy and Management Through Goal-Directed Design and Gamification in Personal Finance Application," *Proceedings of the 27th International Conference on Human-Computer Interaction*, Gothenburg, Sweden, pp. 68-84, June., 2025.

[23] G. M. Santos, R. Julia, and M. Zanchetta, "Personalizing financial literacy learning for children through GenAI and agentic design: A game-based experiment," *Expert Systems with Applications*, vol. 309, art no. 131149, https://doi.org/10.1016/j.eswa.2026.131149, 2026.

[24] E. ter Hoeven, et al., "Mesa 3: Agent-based modeling with Python in 2025," *Journal of Open Source Software*, vol. 10, no. 107, art no. 7668, https://doi.org/10.21105/joss.07668, 2025.

[25] M. Cardinot, C. O' Riordan, J. Griffith, and M. Perc, "Evoplex: A platform for agent-based modeling on networks," *SoftwareX*, vol. 9, pp. 199-204, https://doi.org/10.1016/j.softx.2019.02.009, 2019.

[26] Y. Liu, J. Cao, Z. Li, R. He, and T. Tan, "Breaking Mental Set to Improve Reasoning through Diverse Multi-Agent Debate," *Proceedings of the 13th International Conference on Learning Representations*, Singapore, pp. 24666-24725, Apr., 2025.

[27] L. Alsemgeest, "Talking about money is taboo: Perceptions of financial planning students and implications for the financial planning industry," *Industry and Higher Education*, vol. 30, no. 6, pp. 394-401, https://doi.org/10.1177/0950422216670065, 2016.

[28] R. L. Lamb, L. Annetta, J. Firestone, and E. Etopio, "A meta-analysis with examination of moderators of student cognition, affect, and learning outcomes while using serious educational games, serious games, and simulations," *Computers in Human Behavior*, vol. 80, pp. 158-167, https://doi.org/10.1016/j.chb.2017.10.040, 2018.

[29] L. Lisana, H. Dinata, and G. V. Tanudjaja, "Playing to learn: Game-based approach to financial literacy for generation Z," *Entertainment Computing*, vol. 52, art no. 100896, https://doi.org/10.1016/j.entcom.2024.100896, 2025.

[30] L. Platz and M. Jüttler, "Game-based learning as a gateway for promoting financial literacy – how games in economics influence students' financial interest," *Citizenship, Social and Economics Education*, vol. 21, no. 3, pp. 185-208, https://doi.org/10.1177/14788047221135343, 2022.
[31] F. D' Acunto and A. G. Rossi, "Robo-Advising," *The Palgrave handbook of technological finance*, Palgrave Macmillan, pp. 725-749, https://doi.org/10.1007/978-3-030-65117-6_26, 2021.
[32] T. Hu, Z. Tan, S. Wang, H. Qu, and T. Chen, "Multi-Agent Debate for LLM Judges with Adaptive Stability Detection," Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025), San Diego, CA, USA, pp. 46504-46540, https://doi.org/10.52202/085713-1548, Dec., 2025.
[33] S. Chen, D. Lian, Y. Hong, L. Weng, and D. Hong, "G-DMAD: Group-Based Diverse Multi-Agent Debate for Robust Reasoning," *IEEE Access*, vol. 14, pp. 47289 -47300, https://doi.org/10.1109/ACCESS.2026.3667025, 2026.
[34] N. Chen, Y. Wang, Y. Deng, and J. Li, "The Oscars of AI Theater: A Survey on Role-Playing with Language Models," arXiv:2407.11484, 2024.
[35] M. Kim, B. Kim, and K. Han, "QuerySwitch: Supporting the Design Process by Balancing Vagueness through Large Language Models," *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona, Spain, pp. 1-18, Apr., 2026.
[36] J. Kim, et al., "Laps: Automating Hypothesis-Driven Statistical Analysis of Public Survey Using Large Language Models," *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona, Spain, pp. 1-19, Apr., 2026.
[37] T. Guo, et al., "MASS: Multi-agent simulation scaling for portfolio construction," arXiv:2505.10278, 2025.
[38] J. Wu, D. Choi, B. Kim, and H. Hong, "When Special Education Meets LLMs: Investigating the use of LLM-based Conversational AI by Special Education Teachers for Autism in China," *Proceedings of the 2026 Designing Interactive Systems Conference*, Singapore, pp. 2392-2408, June., 2026.
[39] D. Hooshyar, et al., "Why Large Language Models Alone Fall Short for Responsible Learner Modelling in K-12 Tutoring: A Case Study," *IEEE Access*, vol. 14, pp. 91267-91284, https://doi.org/10.1109/ACCESS.2026.3701047, 2026.
[40] A. Lusardi, O. S. Mitchell, and V. Curto, "Financial Literacy among the Young," *Journal of Consumer Affairs*, vol. 44, no. 2, pp. 358-380, https://doi.org/10.1111/j.1745-6606.2010.01173.x, 2010.
[41] B. Kim, S.-W. Lee, H. Hong, and K. Han, "Automated Time Manager: Effectiveness of Self-Regulation on Time Management Through a Smartphone Application," *IEEE Access*, vol. 7, pp. 90891-90903, https://doi.org/10.1109/ACCESS.2019.2926743, 2019.
[42] C. Gunaratne and I. Garibay, "NL4Py: Agent-based modeling in Python with parallelizable NetLogo workspaces," *SoftwareX*, vol. 16, art no. 100801, https://doi.org/10.1016/j.softx.2021.100801, 2021.
[43] J. Brooke, "SUS: a 'quick and dirty' usability scale," *Usability evaluation in industry*, pp. 189-194, 1996.
[44] S. G. Hart and L. E. Staveland, "Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research," *Advances in Psychology*, Elsevier, vol. 52, pp. 139-183, 1988.
[45] A. Bangor, P. Kortum, and J. Miller, "Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale," *Journal of usability studies*, vol. 4, no. 3, pp. 114-123, https://doi.org/10.5555/2835587.2835589, 2009.
[46] J. R. Landis and G. G. Koch, "The Measurement of Observer Agreement for Categorical Data," *Biometrics*, vol. 33, no. 1, pp. 159-174, https://doi.org/10.2307/2529310, 1977.

저 자 소 개



정다윤(Dayun Jeong)
2025년~현재 : 충북대학교 정보통신공학부 학사과정

관심분야 : AI Agent, Financial Education
ORCID Number : 0009-0001-0124-5590
E-mail : jdy178@cbnu.ac.kr



이동현(Dong Hyeon Lee)
2021년~현재 : 충북대학교 정보통신공학부 학사과정

관심분야 : Human-Computer Interaction, AI Agent, LangChain
ORCID Number : 0009-0005-3326-3196
E-mail : gd091330@chungbuk.ac.kr



최윤선(Yoonseon Choi)
2024년 : 한양대학교 데이터사이언스학부 공학사
2026년 : 한양대학교 데이터사이언스학부 공학석사
2026년~현재 : 한양대학교 연구원

관심분야 : Human Computer Interaction, Time-series Forecasting, Digital Healthcare
ORCID Number : 0009-0002-7797-4260
E-mail : yoonseonchoi@hanyang.ac.kr



강윤석(David Yoon Suk Kang)
2013년 : 한양대학교 공과대학 컴퓨터공학부 공학사
2022년 : 한양대학교 컴퓨터소프트웨어학과 공학박사
2022년~2024년 : 미국 미시건대학교 박사후연구원
2024년~현재 : 충북대학교 소프트웨어학부 조교수

관심분야 : Data Mining, Graph Machine Learning
ORCID Number : 0000-0002-5892-2265
E-mail : dyskang@cbnu.ac.kr



김보관(Bogoan Kim)
2017년 : 미네소타대학교 통계학부 이학사
2025년 : 한양대학교 인텔리전스컴퓨팅학과 공학박사
2025년 : KAIST 장영실 펠로우
2025년~현재 : 충북대학교 정보통신공학부 조교수

관심분야 : Human-Computer Interaction
ORCID Number : 0000-0002-9083-1128
E-mail : bogoan@cbnu.ac.kr